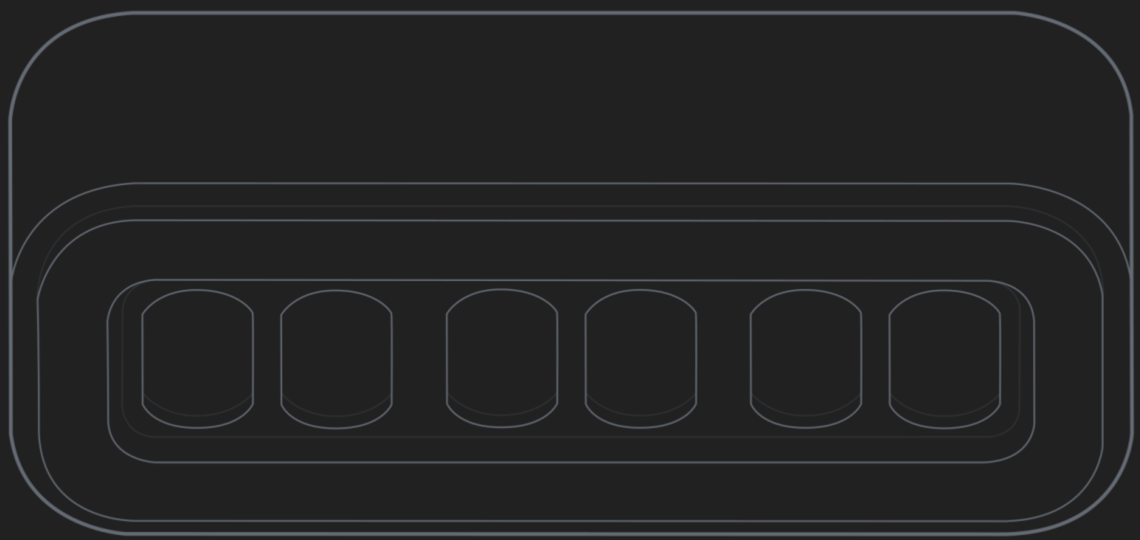
Walnut IN-12 GPS Clock – User Manual



1. User Warnings

- This device contains high-voltage circuitry. Do not open the enclosure.
- Do not operate the clock in wet or humid environments.
- Avoid exposing the clock to extreme temperatures or direct sunlight.
- Use only the provided or recommended power supply.
- If the device is damaged or malfunctioning, discontinue use immediately.

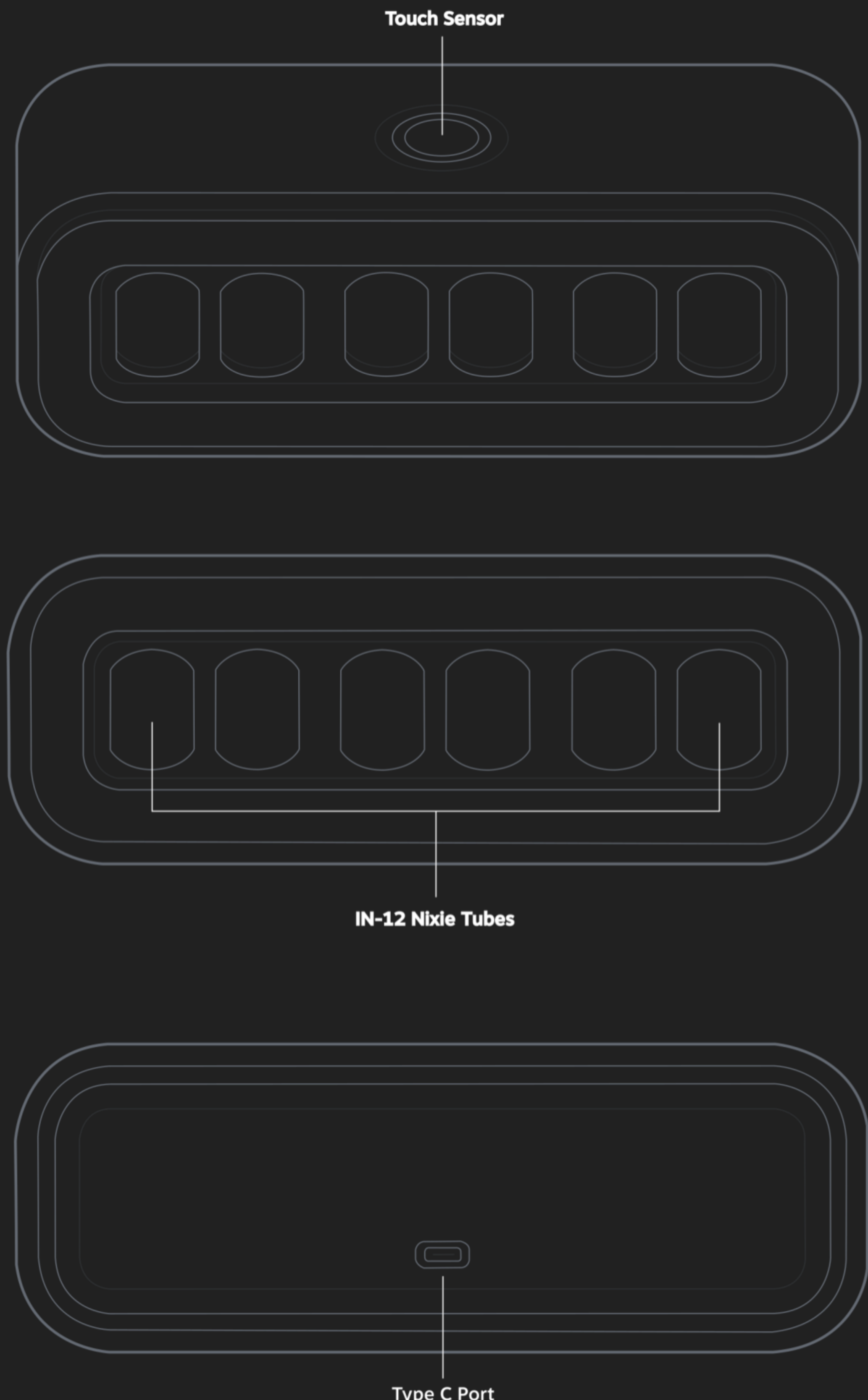
2. What's Included

- 1x Nixie Tube Clock (fully assembled)
- 1x USB-C Power adapter

3. Quick Start Guide

1. Plug in the clock using the provided power adapter.
2. Place the clock near a window or area with good sky visibility.
3. Wait for the GPS module to acquire a signal (typically 5–10 minutes on first use).
4. Once locked, the clock will automatically set the correct time.
5. Use the touch sensor to configure time zone and preferences.

Note: If you plan to use the clock in a basement or low-signal area, perform the initial setup near a window first. After a GPS fix is obtained, the clock can be moved—but it will slowly drift over time without periodic GPS updates.


Touch Sensor
IN-12 Nixie Tubes
Type C Port

4. How the Clock Works

GPS Time Synchronization

- The clock obtains a GPS time fix once every 5 minutes.
- This provides highly accurate atomic time reference.
- On each successful fix, the clock corrects any drift.

Placement Notes

- Best performance is achieved near a window.
- In most homes, the clock can still acquire signal within ~10 ft of a window.
- Basements or heavily obstructed areas typically prevent signal acquisition.

5. Touch Sensor Functions

The clock is controlled using a single touch sensor located on the top center of the device.

Single Quick Press

- Toggle Daylight Saving Time (DST) on/off

Double Quick Press

- Toggle between 12-hour and 24-hour modes

Hold (5 Seconds)

- Enter UTC Offset Setup Mode

In UTC Offset Mode:

- Single tap: Cycle to the next UTC offset
- No input for 5 seconds: Exit and return to normal operation

6. UTC Offset Reference Table

Menu Value	UTC Offset	Menu Value	UTC Offset	Menu Value	UTC Offset
1	UTC-12:00	14	UTC-1:00	27	UTC+7:00
2	UTC-11:00	15	UTC+0:00	28	UTC+8:00
3	UTC-10:00	16	UTC+1:00	29	UTC+8:45
4	UTC-9:30	17	UTC+2:00	30	UTC+9:00
5	UTC-9:00	18	UTC+3:00	31	UTC+9:30
6	UTC-8:00	19	UTC+3:30	32	UTC+10:00
7	UTC-7:00	20	UTC+4:00	33	UTC+10:30
8	UTC-6:00	21	UTC+4:30	34	UTC+11:00
9	UTC-5:00	22	UTC+5:00	35	UTC+12:00
10	UTC-4:00	23	UTC+5:30	36	UTC+12:45
11	UTC-3:30	24	UTC+5:45	37	UTC+13:00
12	UTC-3:00	25	UTC+6:00	38	UTC+14:00
13	UTC-2:00	26	UTC+6:30		

7. System Backup Function

The clock uses a CR1220 battery to preserve settings during power loss.

The following settings are retained:

- UTC offset
- Daylight Saving Time (DST)
- 12/24-hour mode

Note: The display turns off when power is lost, but settings remain saved.

8. Tube Cleaning Cycle

Over time nixie tubes can develop cathode poisoning if digits aren't used frequently, so to maintain performance and longevity, the clock automatically runs a cleaning cycle.

What Happens

- All digits cycle rapidly through 0–9
- Each cathode is evenly exercised
- Normal display resumes afterward
- Cleaning cycle will run every 15 minutes

This is normal operation and not a malfunction.

9. Troubleshooting

Clock Not Setting Time

- Ensure the clock has access to a GPS signal
- Move closer to a window
- Allow several minutes for signal acquisition

Clock Drifting Over Time

- Ensure periodic GPS signal reception
- Relocate temporarily for re-synchronization if needed

Tubes Not Lighting Properly

- Allow cleaning cycle to run
- Check power supply connection

Buttons Not Responding

- Ensure correct press timing (single, double, or hold)
- Restart the device

10. Maintenance & Care

- Clean with a dry microfiber cloth
- Do not use liquids or solvents
- Keep away from dust-heavy environments

11. Technical Specifications

- Display: 6x IN-12 Nixie Tubes
- Dimensions: 9”L x 3.7”W x 3.2”H
- Time Format: HH:MM:SS
- Power Input: PD3.0 Type C power supply, 30W
- Backup Battery: CR1220